

Arun Sharma

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SUMMARY

My research advances Spatial AI and Physical Intelligence by fusing deep generative models, foundation models, and physics-informed reinforcement learning with geographic, geometric, and physical priors for accurate, interpretable, and physically-consistent spatial reasoning. I build physics- and geo-aware diffusion frameworks that embed kinematic, geostatistical, and attributed road-network priors for spatial anomaly detection, climate downscaling, and high-fidelity trajectory generation, and generalize them into Physics-guided World Models via conservation laws and governing PDEs (Navier–Stokes, Maxwell). I extend this Physical-AI agenda to knowledge-guided transfer learning for site-calibrated agroecosystem prediction and to scalable spatial-optimization algorithms grounded in time geography for spatiotemporal joint optimization with geometric primitives and Pareto-optimal solutions to dramatically shrink the monitoring area surfaced to end users. Together, these contributions push toward 3D geometric refinements that adhere to physical laws and more interpretable spatial reasoning for multi-modal endpoint applications across geospatial ML, autonomous driving, 3D reconstruction, Gaussian Splatting, Neural Rendering, VLMs, and RL-based spatial intelligence.

EDUCATION

UNIVERSITY OF MINNESOTA, TWIN CITIES

Doctor of Philosophy (Ph.D.) in Computer Science

Minneapolis, MN

July 2018 - August 2025

Advisor: Prof. Shashi Shekhar and Prof. Vipin Kumar | Spatial Computing Research Group

Thesis Committee: Prof. Vipin Kumar, Prof. Ravi Janardan, Prof. Shashi Shekhar, and Prof. Ying Song

STATE UNIVERSITY OF NEW YORK AT BUFFALO

Master of Science (M.S.) in Computer Science

Buffalo, NY

August 2016 - June 2018

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, Scala, C/C++, Julia, Rust, Go, Shell (Bash), MATLAB, JavaScript (Node.js), TypeScript.

Machine Learning: MLflow, XGBoost, MLlib, LLMs (fine-tuning & inference), VLLMs, RAG, LangChain, VectorDB (FAISS, Chroma, Pinecone), Hugging Face Transformers, LoRA, PEFT, RLHF, OpenAI API, Triton Inference Server, ONNX.

Big Data & Distributed Systems: PySpark, Terraform, Flink, Presto, Airflow, Kafka, RabbitMQ, Delta Lake, Docker, Kubernetes.

ML Frameworks & Deployments: PyTorch, TensorFlow, JAX, ONNX, SageMaker, Lambda, ECS, Bedrock.

WORK EXPERIENCE

ESRI - ENVIRONMENTAL SYSTEMS RESEARCH INSTITUTE

Redlands, CA

Research Scientist Intern - Ph.D.

May 2023 – August 2023

- Designed and implemented a scalable Graph-based Traffic Representation and Association (GTRA) framework for maritime route optimization, leveraging PySpark and GeoAnalytics APIs and reducing query latency by 40%.
- Developed an anomaly detection pipeline integrating Transformer-based models, Evidential Deep Learning (EDL), and AWS SageMaker, improving real-time anomaly classification accuracy from 54.76% to 73.02%.
- Developed an automated model training pipeline and optimized inference for GIS-based AI models on a large-scale maritime dataset by leveraging AWS services (Lambda, ECS, SageMaker Pipelines, Multi-Model Endpoints (MME), Step Functions, SQS, CloudWatch) and implementing model quantization, reducing model retraining time by 35% and API latency by 30%.

UNIVERSITY OF MINNESOTA, TWIN CITIES

Minneapolis, MN

Graduate Research Assistant

August 2018 – August 2025

- Proposed Pi-DPM (Physics-informed Diffusion Probabilistic Model), an encoder-decoder diffusion pairing a transformer Context-Informed Encoder (spatial+temporal attention over neighbor trajectories for fine-scale spatiotemporal synchronous dependencies) with a physics-informed decoder embedding a single-axle Kinematic Bicycle Model (S-KBM) prior and a physics regularizer on velocity, acceleration, heading, curvature, and turning rate; detected GPS-spoofed and AI deep-fake trajectories via threshold-based reconstruction-error anomaly detection, beating SOTA anomaly detectors and generative reconstruction baselines (e.g., DiffWave, DiffTraj, ControlTraj) at 5/10/20% anomaly rates, with strong within- and cross-domain transfer for maritime-safety and homeland-security.
- Designed Kriging-informed Conditional Diffusion Probabilistic Model, a U-Net conditional diffusion conditioned on Universal Kriging interpolations under a Matérn variogram with a variogram-based regularizer penalizing empirical-vs-observed variogram discrepancy in reverse diffusion for fine-scale spatial dependence and physical consistency; downscaled Copernicus CDS and CMIP6 HighResMIP (EC-Earth3P) coarse projections to fine-resolution sea-level anomaly and eddy kinetic energy maps, beating bicubic, CNN, GAN, and baseline-diffusion downscalers on RMSE/MAE/PCC/CRPS across Eastern/Western North America for climate-impact assessment and coastal flooding.
- Built GCDM (Geo-lucid Conditional Diffusion Model), a two-stage cascaded generative diffusion injecting attributed road-network priors via map-informed latent variables, spatial cross-attention, and a physics-based trajectory decomposition (map-derived base + learned residual); generated synthetic vehicle trajectories with stronger geo-distribution similarity (density, trip error) and higher velocity/acceleration/jerk fidelity than DiffTraj, Sashimi, DiffWave, and EETG on Porto and Harbin taxi data, while reducing downstream traffic in/out-flow prediction error.
- Implemented a spatial-variability-aware pretraining-and-fine-tuning extension of KGML-ag and SDSA-KGML using a global-to-local transfer scheme that fine-tunes a globally pretrained physics-guided backbone per site/state via lightweight calibration heads; achieved 17.0%–43.6% relative MSE reductions on Iowa/Illinois/Indiana land-emissions prediction (Ra, Rh, NEE, crop yield) over site-only baselines while preserving KGML-ag mass-balance constraints and local interpretability, with robustness across learning-rate and batch-size perturbations.
- Built MBOR (Multi-Level Bi-Objective Routing), introducing three data structures, a boundary multigraph, a Multi-level Encoded Pareto Frontier View (MEPFV), and 2D cost-interval pruning with multi-edge pruning, to precompute and retrieve complete Pareto-optimal path sets over fragmented road networks; achieved >10× online-runtime improvement for real-time travel-time-vs-energy EV route planning.
- Developed STAGD (Space-Time Aware Gap Detection) with a Dynamic Region Merge (DRM) criterion combining comparison-less temporal indexing, R*-tree hierarchical spatial indexing, and a maximal-union merge over space-time prism geo-ellipses to detect abnormal trajectory gaps (signal-coverage denial, clandestine rendezvous) in large GPS data; proven correct and complete and substantially faster than the baseline on synthetic and MarineCadastre AIS data for maritime-safety applications (illegal fishing, oil transfers, transshipments).

- Proposed TGARD (Time Slicing-based Gap-Aware Rendezvous Detection) and a Dual Convergence DC-TGARD variant exploiting ellipse symmetry with bi-directional pruning and an early-stopping criterion, tightening spatial bounds on possible-rendezvous regions within GPS trajectory gaps over spatial networks vs traditional space-time prism methods; provably correct and complete, experimentally faster and more accurate than ST-prism baselines on synthetic and Geolife Beijing road-network data for homeland-security, and public-health applications.
- Designed a time-geography-based spatiotemporal query optimization framework for trajectory-gap rendezvous detection leveraging space-time paths, prisms, and stations for temporal slicing, reachability constraints, and early pruning to reduce candidate explosion in spatiotemporal joins; introduced a Time Prioritizer (alibi-query ST-prism intersection) and two spatial filters, a Static Ellipse Intersection Filter exploiting prism elliptic-projection geometry and a Dynamic Circle Intersection Filter using polynomial circle bounds per time slice, tightening grid-cell comparisons over a 3D R-Tree / MOBR baseline; proven correct and complete with $O(|K|^2|N|^2)$ complexity, beating baselines on area pruning effectiveness, execution efficiency, accuracy, and precision on synthetic and MarineCadastre AIS data (150K ships, 2009–2017) for homeland-security (illegal fishing, oil transfers, trans-shipments), epidemiology contact tracing, and public-safety use cases.

SELECTED PUBLICATIONS

[1] Towards Physics-informed Diffusion for Anomaly Detection in Trajectories: A Summary of Results
 Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection (GeoAnomalies '25)
Arun Sharma, Mingzhou Yang, Majid Farhadloo, Subhankar Ghosh, Bharat Jayaprakash, and Shashi Shekhar

[2] Towards Kriging-informed Conditional Diffusion for Regional Sea-Level Data Downscaling: A Summary of Results
 32nd International Conference on Advances in Geographic Information Systems, 2024
 Authors: S Ghosh*, **Arun Sharma***, J. Gupta, A. Subramanian and Shashi Shekhar
 (*Both authors contributed equally to this paper)

[3] Physics-based Abnormal Trajectory-Gap Detection
 ACM Transactions in Intelligent Systems and Technology, 2024.
Arun Sharma, Subhankar Ghosh, and Shashi Shekhar

[4] Analyzing Trajectory Gaps for Possible Rendezvous Regions
 ACM Transactions in Intelligent Systems and Technology, 2022
Arun Sharma and Shashi Shekhar

[5] Towards a Tighter Bound on Possible-Rendezvous Areas: Preliminary Results
 30th International Conference on Advances in Geographic Information Systems, 2022
Arun Sharma, Jayant Gupta, Subhankar Ghosh, and Shashi Shekhar

[6] Geo-lucid Conditional Diffusion Models for High Physical Fidelity Trajectory Generation
 Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)
 Mingzhou Yang, **Arun Sharma**, Majid Farhadloo, Bharat Jayaprakash, and Shashi Shekhar

[7] Towards Surrogate Models with Hybrid Spatial Neural Networks: A Summary of Results
 Proceedings of the 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25), pp. 57–69, 2025
 Shengya Zhang, **Arun Sharma**, Majid Farhadloo, Mingzhou Yang, Yao Zhang, Mu Hong, Licheng Liu, David Mulla, and Shashi Shekhar

[8] Towards Fine-Tuning-Based Site Calibration for Knowledge-Guided Machine Learning: A Summary of Results
 5th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)
 Ruolei Zeng, **Arun Sharma**, Shuai An, Mingzhou Yang, Shengya Zhang, Licheng Liu, David Mulla, and Shashi Shekhar

[9] Spatially-Delineated Domain-Adapted AI Classification: An Application for Oncology Data
 SIAM International Conference on Data Mining, 2025
 M Farhadloo, **Arun Sharma**, A. Leontovich, S N. Markovic, and Shashi Shekhar

[10] Towards Spatially-Lucid AI Classification in Non-Euclidean Space: An Application for MxIF Oncology Data
 SIAM International Conference on Data Mining, 2024.
 M Farhadloo, **Arun Sharma**, J. Gupta, A. Leontovich, S N. Markovic, and Shashi Shekhar

[11] Spatial Computing Opportunities in Biomedical Decision Support: The Atlas-EHR Vision
 ACM Transactions on Spatial Algorithms and Systems 10, no. 3 (2024): 1-36.
 M Farhadloo, **Arun Sharma**, A. Leontovich, S N. Markovic, and Shashi Shekhar

HONORS AND ACHIEVEMENTS

DOCTORAL DISSERTATION FELLOWSHIP	2022 - 2023
University of Minnesota, Twin Cities	

REVIEWER

NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, IJCAI, SIGKDD, CIKM, ICDM, SDM, MLSys, SIGSPATIAL, TKDE, JMLR

REFERENCES (CONTACT INFORMATION)

Prof. Shashi Shekhar 5-203 Kenneth H. Keller Hall 200 Union Street SE Minneapolis, MN 55455 Email: shekhar@umn.edu Phone: 612-624-8307 Personal Website	Prof. Vipin Kumar 5-225C Kenneth H. Keller Hall 200 Union Street SE Minneapolis, MN 55455 Email: kumar001@umn.edu Phone: 612-624-8023 Personal Website	Prof. Ying Song 548 Social Sciences Building 267 19th Ave S Minneapolis, MN 55455 Email: yingsong@umn.edu Phone: 612-625-1112 Personal Website	Prof. Varun Chandola 212 Capen Hall (CARA Suite), University at Buffalo, Buffalo, NY 14260 Email: chandola@buffalo.edu Phone: (716) 645-4747 Personal Website
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